

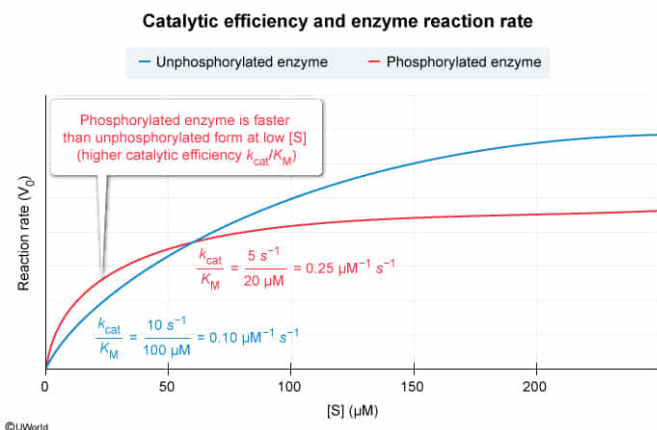
Enzyme A catalyzes the conversion of substrate S to product P. When enzyme A is phosphorylated, its k_{cat} value changes from 10 s^{-1} to 5 s^{-1} and its K_{M} value changes from $100 \text{ }\mu\text{M}$ to $20 \text{ }\mu\text{M}$. If equal concentrations of each enzyme form are used then compared to the unphosphorylated enzyme A, the phosphorylated form:

- ✔ ☒ A. has a higher catalytic efficiency and yields a faster reaction rate when $[\text{S}] < 20 \text{ }\mu\text{M}$. [49%]
- ☐ B. has a higher turnover number and yields a faster reaction rate under saturating conditions. [25%]
- ☐ C. has a lower affinity for S and yields a slower reaction rate at all $[\text{S}]$. [12%]
- ☐ D. is allosterically inhibited and yields a lower reaction rate at all $[\text{S}]$. [12%]

Correct

50% answered correctly

Explanation:



Catalytic efficiency is a measure of how quickly an enzyme catalyzes a reaction at low substrate concentrations (eg, $[\text{S}] < K_{\text{M}}$). An **enzyme is efficient** if a **small amount of substrate** yields a **high reaction rate** V_0 .

Catalytic efficiency is calculated as the ratio of the **turnover number** k_{cat} to the **Michaelis constant** K_{M} (ie, **catalytic efficiency** = $k_{\text{cat}}/K_{\text{M}}$). A higher k_{cat} and/or a lower K_{M} contribute to a higher catalytic efficiency, and therefore to a faster reaction rate at low substrate concentrations relative to less-efficient

enzymes.

The question indicates that enzyme A undergoes a decrease in both k_{cat} and K_{M} upon **phosphorylation**. However, K_{M} decreases by a greater factor than k_{cat} , so the $k_{\text{cat}}/K_{\text{M}}$ ratio *increases*. Therefore, the phosphorylated form of the enzyme has a **higher catalytic efficiency** than the unphosphorylated form, and the phosphorylated form yields a **faster reaction rate at low concentrations** (eg, below 20 μM).

This result can be verified by plugging the relevant numbers into the **Michaelis-Menten equation**. In this equation, V_{max} can be replaced with $k_{\text{cat}} \times [\text{enzyme}]$ (where [enzyme] may be set to any desired level, eg, 1 nM), and substrate is set to any value where $[S] < 20 \mu\text{M}$ (eg, 10 μM):

$$V_0 = \frac{\text{Unphosphorylated enzyme}}{10 \text{ s}^{-1} \times 1 \text{ nM} \times 10 \mu\text{M}} = 0.9 \text{ nM/s} \quad V_0 = \frac{\text{Phosphorylated enzyme}}{5 \text{ s}^{-1} \times 1 \text{ nM} \times 10 \mu\text{M}} = 1.7 \text{ nM/s}$$

(Choice B) The turnover number (k_{cat}) is *lower* for the phosphorylated enzyme, not higher. This results in a **lower reaction rate** than the unphosphorylated form under **saturating conditions**.

(Choice C) The lower K_{M} in the phosphorylated state indicates a **higher affinity** for substrate. Whether the phosphorylated form catalyzes a faster or slower reaction than the unphosphorylated form depends on $[S]$.

(Choice D) Phosphorylation of an enzyme is a **post-translational modification**, not an allosteric event. **Allostery** occurs when a ligand *noncovalently* binds a site on the enzyme other than the active site whereas phosphorylation occurs through a covalent bond.

Educational objective:

Catalytic efficiency is the ratio of k_{cat} to K_{M} for an enzyme and measures how effective an enzyme is at low substrate concentrations. More efficient enzymes have high k_{cat} and low K_{M} values and require only a small amount of substrate to achieve a significant reaction rate.

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